



Article

Investigation of Dynamic Behavior of Confined Ionic Liquid [BMIM]⁺[TCM][−] in Silica Material SBA-15 Using NMR

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Abstract: The molecular dynamics of 1-butyl-3-methyl imidazolium tricyanomethanide ionic liquid [BMIM]⁺[TCM][−] confined in SBA-15 mesoporous silica were examined using ¹H NMR spin-lattice (T₁) relaxation and diffusion measurements. An extensive temperature range (100 K–400 K) was considered in order to study both the liquid and glassy states. The hydrogen dynamics in the two states and the self-diffusion coefficients of the cation [BMIM]⁺ above the glass transition temperature were extracted from the experimental data. The results were then compared to the corresponding bulk substance. The effects of confinement on the dynamic properties of the ionic liquid clearly manifest themselves in both temperature regimes. In the high-temperature liquid state, the mobility of the confined cations reduces significantly compared to the bulk; interestingly, confinement drives the ionic liquid to the glassy state at a higher temperature T_g than the bulk ionic liquid, whereas an unusual T₁ temperature dependence is observed in the high-temperature regime, assigned to the interaction of the ionic liquid with the silica-OH species.

Keywords: confined ionic liquids; NMR; relaxation and diffusion; mesoporous silica SBA-15



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1. Introduction

Water and several other liquids have been studied under geometrical confinement for decades, and their behavior can be considered reasonably well understood [1]. Confined liquids are subject to two concurring effects: the presence of the surface and the geometrical aspect of the restricted space itself [2]. While the latter affects thermodynamic properties, such as melting point and glass transition temperatures, the former possesses the potential, depending on the nature of the solid surface, to enhance or reduce molecular order and restrict or even promote mobility on a molecular scale [3].

Room temperature ionic liquids (ILs) are a class of organic salts that consists of a variety of cations and anions. ILs are liquids at room temperature with fascinating properties depending on the type of cation or anion used to make them. They have unique characteristics that make them attractive in many applications ranging from energy storage and production devices, such as batteries [3,4] and super capacitors [5], to gas absorption systems [6]. ILs are used as bulk or incorporated within materials with a high surface to volume ratio, such as porous materials. The ordering and mobility of ILs within (nano)pores show significant enhancement in catalytic activity [7], charge transport, and electrical conductivity [8,9]. Therefore, immobilization of ILs in solid porous materials renders them more suitable for a broad range of applications.

It has been proposed that the confinement changes the cation–anion distances, which in turn affect their properties [10]. Therefore, understanding the transport properties as well as the ionic and solvent interactions of ILs enables the prediction of their physicochemical

properties. The impact of structure, molecular organization, and cation–anion interactions on the physical properties of ILs is still a topic of major research interest [1,11]. Furthermore, many ILs belong to the class of glass-forming materials, exhibiting a transition from the liquid to a rigid glassy phase at a transition temperature T_g , whose microscopic origin of the temperature dependence of structural and dynamic properties is still not well-understood [12–14].

Under this context, a great number of experimental and theoretical efforts have been employed in order to examine guest–host molecular systems [15–17], including structural relaxation times and transport coefficients of confined glass-forming materials [18]. From the experimental point of view, the dynamical properties of many ILs have been investigated with several techniques, such as dielectric spectroscopy [19,20], neutron scattering [21–24], light scattering [19], and NMR spectroscopy [17,25–32]. Theoretical studies using Molecular Dynamics (MD) simulations also reveal complex phase transitions of nano-confined ILs [32–34]. We refer readers to Ref. [35], which includes detailed information on various experimental studies dealing with the changes in the physicochemical properties (such as thermal phase transition, stability, dynamical behavior, and optical properties) of ionic liquids confined inside porous materials.

In a recent publication [32], we combined MD simulations with ^1H NMR and showed that $[\text{BMIM}]^+[\text{TCM}]^-$ entrapped into the narrow MCM-41 silica nanopores exhibited a complex dynamic ordering and stratified diffusion close to single file diffusion in the liquid state high-temperature regime. This effect was not observed in the wider SBA-15 nanoporous silica, where almost normal diffusion was detected.

It should be stressed that many studies have investigated the dynamics of ILs in confined geometries of silica materials [36–38], carbon nanotubes [39], and porous polymer monoliths [40]. However, most of them (with few exceptions using mainly dielectric relaxation spectroscopy [41,42]) examined the IL mobility within the high-temperature phase above the glass transition temperature. This is due to the fact that all translation degrees of freedom of the ILs freeze out at low temperatures within the glassy state, making the macroscopic observables (self-diffusion coefficients and viscosity) difficult to measure.

In this work, we report ^1H NMR relaxation and 2D NMR diffusion measurements of IL (1-butyl-3-methyl imidazolium tricyanomethanide ionic liquid $[\text{BMIM}]^+[\text{TCM}]^-$) in bulk as well as confined in SBA-15 (average pore size of 6–11 nm) mesoporous materials. The measurements cover a wide range of temperatures (100 K–400 K) to obtain information on the molecular dynamics of this IL in both the liquid and the glassy states (above and below the glass transition temperatures for bulk and confined IL [43]).

2. Results and Discussion

2.1. T_1 Relaxation Time Measurements

Figure 1 shows the structure of a cluster of $[\text{BMIM}]^+[\text{TCM}]^-$ IL molecules used in this study (with their geometry optimized using ORCA software). As can be observed, the $[\text{BMIM}]^+$ cation contains fifteen ^1H nuclei of which six belong to the two methyl groups. One is attached to the end of the alkyl chain, and the other is attached to the imidazolium of the cation. These groups contribute to approximately half (or specifically 6/15) of the ^1H NMR signal, both for the bulk IL and the IL confined inside SBA-15.

NMR relaxation times are determined by the fluctuations of the local magnetic fields inside the material, which are modulated by the molecular motions activated by the thermal energy of the molecules. Therefore, NMR spin-lattice relaxation time T_1 or spin-spin relaxation time T_2 are important in order to acquire direct information on the molecular dynamics. For example, observation of an NMR T_1 minimum on the T_1 versus temperature curve gives a direct measurement of the characteristic correlation time τ_c of the molecular motion through the well-known NMR relationship $\omega_L \tau_c \sim 1$, where ω_L is the Larmor angular frequency for the active NMR nuclei [46], in this case, ^1H .

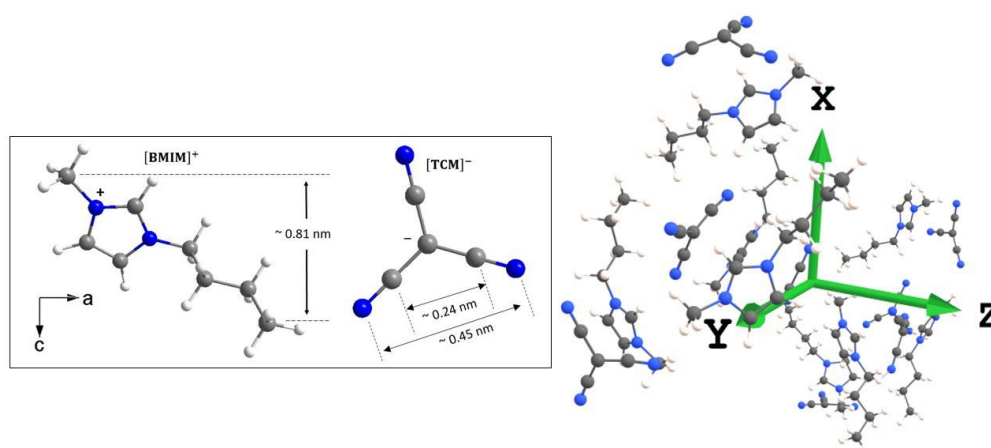


Figure 1. Left: A single molecule of the ionic liquid [BMIM]⁺[TCM][−] used in this study. The [BMIM]⁺ cation is the 1-butyl-3-methylimidazolium (C₈H₁₅N₂) molecule, and the [TCM][−] is the tricyanomethanide (C₄N₃) molecule. Right: The molecular arrangement of a cluster of IL molecules optimized using the ORCA software package [44,45]. Blue, grey, and white circles are nitrogen, carbon, and hydrogen atoms, respectively.

Figure 2 shows the ¹H spin-lattice T_1 relaxation time of the bulk and confined [BMIM]⁺[TCM][−] IL as a function of temperature. Region I and region II represent the temperature ranges above and below the glass transition temperature for bulk IL [43]. The glass transition temperatures for the bulk and confined ionic liquids were measured using differential scanning experiments (Appendix A, Figure A1), where T_g for bulk IL is 186.4 K, and T_g for confined IL inside SBA-15 is 190.5 K. The spin-relaxation times versus reciprocal temperature are also shown in the Appendix A, Figure A2.

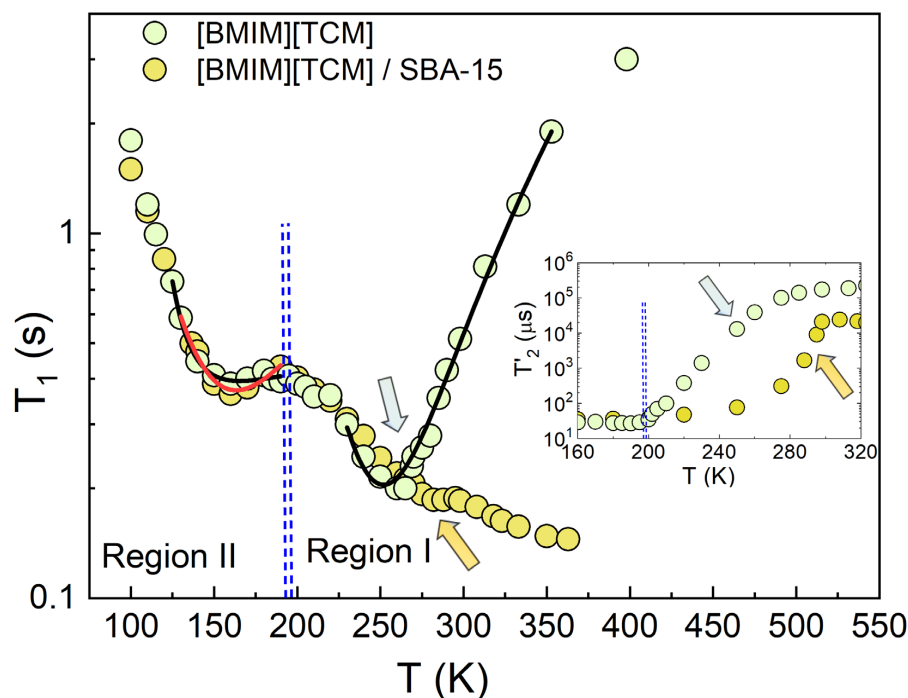


Figure 2. Spin-lattice T_1 relaxation time as a function of temperature for the bulk IL and confined IL in SBA-15. The two vertical dashed lines are the glass transition temperatures for the bulk ($T_g = 186.4$ K) and confined ILs ($T_g = 190.5$ K), obtained using the DSC technique. The arrows indicate the temperatures at which T_1 minima occur above T_g (Region I). The solid curves are the theoretical fits to the experimental data using the BPP model (refer to the text). The inset of the figure shows the spin-spin relaxation time T_2 versus temperature.

At the low-temperature region, T_1 minima are observed for both liquids at ~160 K. In addition, the data show a well-defined T_1 minimum at high T for the bulk IL and a less pronounced T_1 minimum for the confined IL (shown with arrows in Figure 2). The inset of the figure shows the spin-spin T_2 relaxation time versus temperature. The decrease in the T_2 s for both the bulk and confined ILs reflects the freezing of the translational motion; remarkably, in the case of the confined IL, freezing starts at a sufficiently higher temperature than the bulk IL.

It is important to note that the OH protons on the surface of the silica materials have long T_1 (4.0 s–5.0 s) and very short T_2 (40 μ s–50 μ s). Therefore, in our relaxation measurements, precautions were taken to filter out the proton signal from the silica surface OH groups by using appropriate pulse repetition times and Hahn SE interpulse times, particularly at high temperatures where the relaxation times for IL and surface OH protons are significantly different from each other. Besides, the OH proton NMR signal is very weak, as confirmed in Figures 2 and 4 of our recently published work [32], and therefore, it is not expected to influence the ^1H NMR measurements at low temperatures, where T_1 and T_2 values of OH are of the same order of magnitude as those of the ILs. This fact is supported by the similarity of the T_1 vs. T curves at low temperatures for both the bulk IL (where there are no surface OH protons) and the confined IL/SBA-15.

On the basis of the relationship $\omega_L \tau_c \sim 1$, the correlation times τ_c of the bulk and confined ILs are in the sub-microsecond time scale. Nevertheless, a careful analysis of the experimental data is required for a more accurate characterization of the molecular dynamics. One model that is routinely used in fitting T_1 versus temperature data is the model proposed by Bloembergen, Purcell, and Pound [47] (BPP). In this model, the spin-lattice relaxation rate $1/T_1$ depends on the spectral densities $J(\omega_L)$ and $J(2\omega_L)$ of the local magnetic fluctuations at the Larmor frequency, ω_L , through the following formula:

$$\frac{1}{T_1} = 2K \frac{\gamma^4}{r^6} [J(\omega_L) + 4J(2\omega_L)]$$

$$\frac{1}{T_1} = \frac{3}{10} \frac{\hbar^2 \gamma^4}{r^6} \left[\frac{\tau_c}{1 + \omega_L^2 \tau_c^2} + \frac{4\tau_c}{1 + 4\omega_L^2 \tau_c^2} \right]$$

Here, τ_c denotes the correlation time characteristic of the random motion, $\hbar$ is the reduced Planck's constant, γ is the gyromagnetic ratio, and r is the distance between two interacting dipoles.

The ^1H relaxation mechanism in bulk IL at high temperatures, where diffusion is rapid, is controlled by the diffusional modulation of the inter-nuclear dipolar fields as proposed in the BPP model. Consequently, the bulk IL T_1 data in region I were fitted to the BPP model (solid black line shown in Figure 2) assuming that the diffusion process is thermally activated, i.e., $\tau = \tau_0 \exp(E_a/k_B T)$ where E_a is the activation energy for the tumbling motion of molecules under investigation, and τ_0 is the characteristic time of the molecular motion. We refer to the molecular motion as tumbling because, at high temperatures, it is expected that both translational and rotational processes are coupled together. The solid curve in region I of Figure 2 was obtained by using the most probable value for τ_c , which is in good agreement with previously published values in similar [BMIM] $^+$ based ionic liquids [12,21,30]. The average correlation time τ_c at the T_1 minimum at 250 K was calculated to be 1.0 ns.

It is worth noting that, above 260 K, the temperature dependence of T_1 for the bulk IL is different from that in the confined IL. In bulk IL, the T_1 values increase with increasing temperature, while the opposite change is observed for the T_1 data of the confined ILs. The reason is that, at high temperatures, the molecules of bulk IL are moving rapidly with short correlation times where $\tau_c \ll 1/\omega_L$. In this case, there are some limited spectral densities at ω_L and $2\omega_L$ with their low amplitudes stemming from the fact that the energy of the molecular motion is spread over a very broad frequency range. As a result, relaxation times increase with temperature (as observed from the data of bulk IL), in line with what is

expected by BPP theory. Additionally, considering spin-spin interaction (intramolecular and intermolecular interactions) among all the spin pairs, the relaxation is expected to vary with self-diffusion coefficients D as $1/T_{1Bulk} \propto 1/D$ [48] with the explicit equation given as:

$$\left(\frac{1}{T_1}\right)_{translation/Bulk} = \frac{\pi N \gamma^4 \hbar^2}{5 a D} \propto \frac{1}{D}$$

where a represents the molecular size, $\hbar$ is the reduced Plank's constant, and γ is the gyromagnetic ratio of the spins. As we will observe later (Figure 5), at high temperatures and bulk IL, the D is proportional to T , similar to T_1 as observed in Figure 2.

On the other hand, the pore walls and the surface interactions affect the relaxation mechanism for confined molecules inside the silica material. In this case, the relaxation rate is considered to be proportional to the self-diffusion coefficient of the molecules, $1/T_{1wall} \propto D$ [49]. This is because the increase in parameter D , as a response to increasing temperature, enables the molecules to diffuse faster and reach the surface of the pores more often. These molecules then quickly relax due to the presence of unpaired electrons from the oxygen groups on the SBA-15 silica walls [50] (the density of Si-O-OH bonded OH groups on the surface of SBA-15 was reported to be approximately 3.7 OH/nm² [51]). This scenario corroborates with the fact that a layer of adsorbed IL molecules on the surface of the SBA-15 at room temperature decreases gradually with increasing temperature [32], as also shown later in Figure 4.

The T_1 minimum at the low-temperature regime (region II, i.e., below T_g , refer to Figure 2) indicates a type of molecular motion with frequency in the MHz region within the glassy state. We attribute this to the motion of the methyl groups of the IL. It is well known that methyl rotation is the fastest dynamic in organic systems. Therefore, it is expected that this rotation mechanism manifests itself within the low-temperature glassy state. The T_1 minimum in the low-temperature region II can thus be ascribed to the three-fold rotation of the two methyl groups with respect to their C-C bond, where the characteristic correlation time of this motion is comparable to the NMR Larmor period.

Furthermore, ¹H NMR spin-spin relaxation time measurements in the low-temperature regime of region II (inset of Figure 2 and ref. [32]) appear to be temperature independent. This is because, at low temperatures, all the translational motions of the IL cations are frozen out. The only motion surviving is the reorientation of the methyl groups in agreement with previously reported results using NMR techniques in polymers and organic compounds (refer to, for example, Refs. [52–55]).

The experimental T_1 data for bulk IL at low temperatures between 130 K to 190 K were fitted to the BPP model (Figure 2), giving an activation energy E_a for the methyl hindered rotation of approximately 5.0 kJ/mol. The low value of the E_a of 5.0 kJ/mol in region II compared to 22 kJ/mol in region I for bulk IL signifies the difference in the origin of the molecular motion of the IL in the low- and the high-temperature regimes. It is frequently observed that the activation energy for methyl-group rotation, determined from proton T_1 measurements, changes to a lower value as the temperature is reduced [56]. In the high-temperature phase, the motion of the whole molecule requires higher activation energy than the one needed for the rotational motion of methyl groups at the low-temperature regime (refer to, for example, Refs. [57,58]). In addition, in the glassy state, the methyl groups populate different local environments due to the structural disorder inherent to the glassy state. This manifests itself in different rates of rotation [59]. The activation energies E_a for the methyl rotations of the confined ILs at low temperatures (region II) were also calculated using the Arrhenius activation form and were found to be equal to 5.8 kJ/mol, close to the value found for the bulk IL (5.0 kJ/mol). This indicates that the methyl rotation is not significantly perturbed by the confinement effect. Finally, both factors (τ_o , T) contribute to the higher activation energy of the molecular motion in region I compared to that in region II. This is expected because, at the high-temperature region I, the whole molecules are in motion, and therefore, more energy is required to activate motions in comparison to

the low-temperature region II where only methyl group motion persists (which, of course, requires a smaller activation energy).

2.2. Self-Diffusion in the High-Temperature Regime

It is well known that experiments to measure the self-diffusion coefficient D using NMR spectroscopy in the presence of a static or pulsed magnetic field gradient can directly determine the mobility at the atomic and molecular levels [60]. The diffusion coefficients obtained in this way do not depend on any assumptions about microscopic details of the atomic motion, such as jump length or hard-core dimensionality (e.g., Einstein–Smolouchouski’s law [61]); therefore, the measurements are purely hydrodynamic in nature.

In the conventional 1D NMR diffusion experiments with a uniform diffusion process, D is obtained by appropriately fitting the ^1H NMR spin-echo decay data [62]. However, in the case of non-uniform diffusion processes (e.g., liquids confined in pore structures), diffusion is expressed with a distribution function $f(D)$. This function is obtained by implementing an appropriate 1D inversion algorithm on the spin-echo decay data [63].

Alternatively, 2D diffusion-relaxation D - $T_{2\text{eff}}$ correlation spectroscopy is advantageous compared to 1D NMR diffusion experiments, as it allows the measurement of spin-spin relaxation times ($T_{2\text{eff}}$) with the relevant D values [64]. Here, the D - $T_{2\text{eff}}$ technique has been employed to examine the relevant differences between the $T_{2\text{eff}}$ and D values of the bulk and confined ILs. Figure 3 shows the representative 2D contour results for the bulk and confined ILs at 292 K.

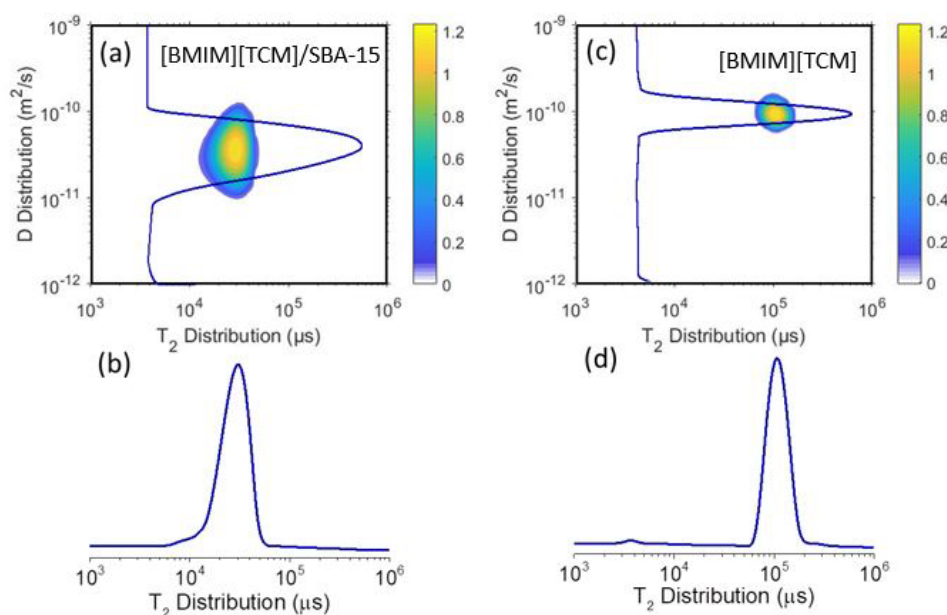


Figure 3. (a,c) 2D ^1H NMR D - $T_{2\text{eff}}$ contour plots of the bulk ionic liquid and confined [BMIM] $^+$ [TCM] $^-$ ionic liquid in SBA-15 at 292 K. (b,d) Distribution of the $T_{2\text{eff}}$ values obtained from the 2D contour plots.

The result for the bulk IL shows a narrow region on the 2D plot represented by a single D value of $\sim 1 \times 10^{-10} \text{ m}^2/\text{s}$ and a single T_2 value of $\sim 8 \times 10^4 \text{ }\mu\text{s}$ at 292 K. Narrow peaks in D and T_2 windows are expected for bulk liquids. On the other hand, the data of the confined IL inside SBA-15 show a remarkably broader distribution of the D values and a relatively broader T_2 distribution, as observed in the 2D plots in Figure 3. The broadening of the D peak for the confined IL inside SBA-15 compared to the bulk IL is mainly because the molecules close to the pore walls are getting a slower diffusion, whilst those at the pore center acquire almost bulk-like diffusion.

This is best observed in Figure 4, which presents the ^1H NMR Hahn spin-echo decays (SEDs) of the two samples at the Larmor frequency of 100 MHz and in a constant magnetic field gradient of $G = 34.7$ Tesla/m at RT and 400 K.

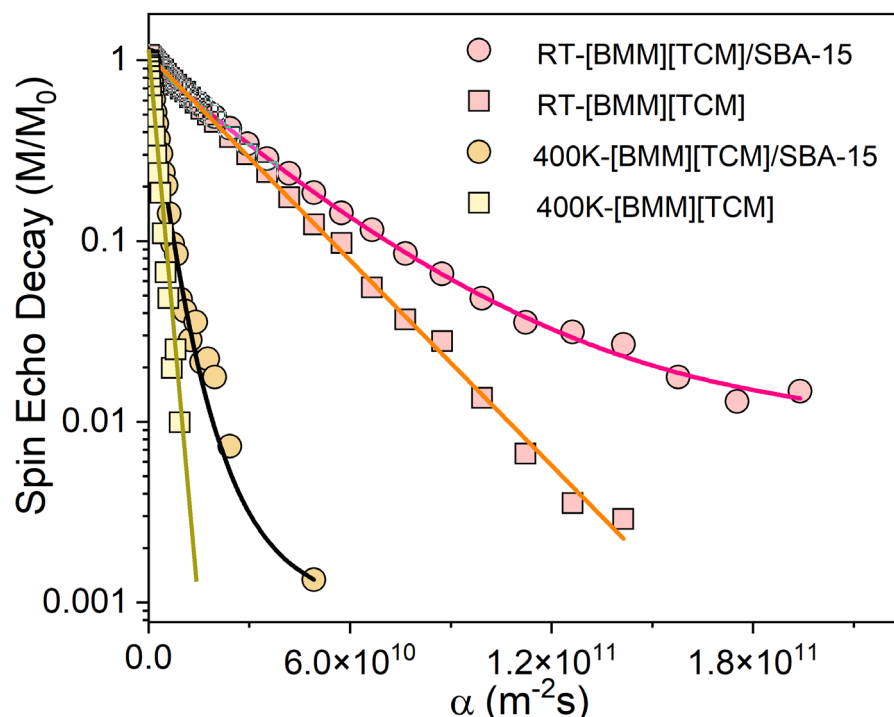


Figure 4. ^1H NMR SED plots of the bulk ionic liquid and confined $[\text{BMIM}]^+[\text{TCM}]^-$ ionic liquid in SBA-15 at room temperature and 400 K at the Larmor frequency of 100 MHz and in a constant magnetic field gradient of $G = 34.7$ Tesla/m. In the case of a homogeneous diffusion process with a single self-diffusion coefficient D , the spin echoes decay according to the law $M/M_0 = \exp(-D\alpha)$, where $\alpha = (2/3)\gamma^2 G^2 \tau^3$.

In the case of a homogeneous diffusion process with a single self-diffusion coefficient D , SEDs decay according to the law $M/M_0 = \exp(-D\alpha)$, where $\alpha = (2/3)\gamma^2 G^2 \tau^3$. Here γ is the gyromagnetic ratio of ^1H . Indeed, in the case of the bulk IL, a single D value is obtained with the values $D = 6 \times 10^{-11} \text{ m}^2/\text{s}$ at RT and 5×10^{-10} at 400 K. However, in the case of the IL inside the SBA-15 nanopores, the slope deviates to lower values as α gets larger, showcasing the presence of a distribution of D -values, which decrease as the IL gets closer to the silica walls, in accordance with Figure 3a. Notably, by comparing the slopes of the bulk and confined ILs, it is observed that, at room temperature, almost 50% of the confined IL behaves as bulk IL (they have the same slope, i.e., D values), while at 400 K, the bulk-like part increases to 80%, indicating that the adsorbed slow-diffusing IL layer becomes thinner with increasing temperature.

In order to study the evolution of the diffusion dynamics in the high-temperature regime, D - T_{eff} experiments were performed in the temperature range 240 K–400 K, and the average log-mean D values at each temperature were extracted, as shown in Figure 5, exhibiting the temperature dependence of the relevant D values.

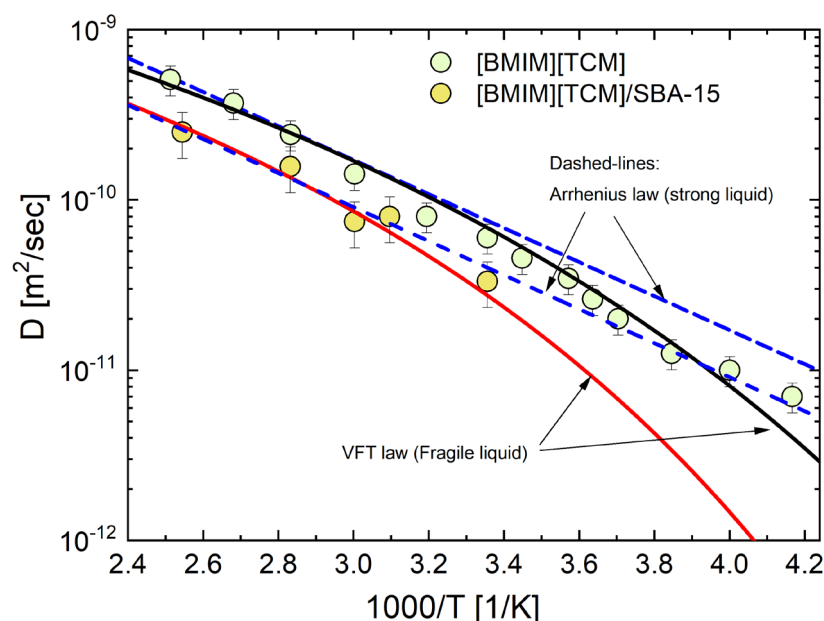


Figure 5. Self-diffusion coefficients D as a function of the inverse temperature obtained from ^1H NMR measurements for the bulk ionic liquid and confined $[\text{BMIM}]^+[\text{TCM}]^-$ ionic liquid in SBA-15. The black and red lines are the fit curves obtained using the VFT law. The dashed lines are the relevant Arrhenius curves.

In the literature, different models have been considered for the data analysis using the concept of fragility introduced by Angel [65], who classified a liquid as a strong liquid or a fragile liquid. This classification is based on whether the liquid's temperature dependence of a dynamic observable, such as self-diffusion coefficients or structural relaxation times, follows an Arrhenius or a non-Arrhenius behavior. A pure Arrhenius behavior classifies a strong liquid, whereas a non-Arrhenius one signifies a fragile behavior. For example, liquid water is a fragile liquid because the temperature dependence of its thermodynamic properties does not obey the Arrhenius law.

For the IL considered in this study, it appears that D varies in a non-Arrhenius way with temperature, at least for the bulk IL, indicating a fragile behavior. This is further observed by comparing the statistical parameters obtained from the fittings, shown in Table A1 in Appendix A. Therefore, we have used a Vogel–Fulcher–Tammann (VFT) law $D = D_0 \exp\left[\frac{-B}{(T-T_0)}\right]$ to fit the diffusion data [66]. The solid black and red curves are the best fit curves using the VFT law, shown in Figure 5. The dashed lines depict the relevant Arrhenius fits, while it is evident that the bulk IL data do not conform to the curve. In the VFT law, the parameter B represents the strength of the activated process, which governs the flow behavior of the material. A lower value of B indicates that the glassy liquid is more fragile in nature. The VFT law diverges at the finite temperature T_0 , where the configurational entropy vanishes [67]. Similar values of B and T_0 were estimated for the bulk IL (810 K, 137 K) and the confined IL/SBA-15 (800 K, 157 K) systems. The adjustable parameters are in agreement with reported values for the $[\text{BMIM}]$ cation [25].

In the case of the confined IL, both fitted curves (Arrhenius and VFT) are substantially in line with the data, yet it appears that the VFT law provides a better representation (refer to Table A1, Appendix A). The reason for this is that the temperature range examined herein is restricted, primarily due to the freezing of the confined IL at relatively high temperatures. Therefore, in this case, no conclusive inference can be drawn on the fragility of the confined liquid. Further research is required in this regard.

Finally, it is observed that, in the high-temperature liquid state, the mobility of the confined cations reduces significantly compared to that of the bulk IL as revealed by the

lower values of the self-diffusion coefficients. This is in accord with results reported in the literature [31,68].

3. Materials and Methods

3.1. Chemicals and Porous Materials

The [BMIM]⁺[TCM][−] (mass fraction purity >98% with 720 ppm H₂O content) was purchased from IoLiTech GmbH. Mesoporous SBA-15 with cylindrically shaped pores [69] (nominal pore size 6–11 nm) was obtained from ACS Material. Prior to sample preparations, the silica powder was dried overnight at 100 °C under a weak pump system to remove any possible water vapor within the material. The procedure for pore filling with the IL was previously reported in ref. [32].

3.2. NMR Spectroscopy

NMR relaxation experiments were performed at 100.123 MHz using a broadband pulse NMR spectrometer. For the spin-lattice relaxation time T_1 measurements, the typical NMR inversion recovery pulse sequence (180°- τ -90°-signal-) was used. The 90° pulse-width was set to 3.0 μ s; the inter-pulse duration was 10–15 μ s with a repetition time of 6.0 s. For the spin-spin relaxation time T_2 measurements, the Hahn spin-echo (SE) pulse sequence was used, 90°- τ -180°- τ -SE (90° pulse width was set to 3.0 μ s). 2D NMR diffusion-relaxation D - $T_{2\text{eff}}$ experiments were conducted in the fringe field of a 4.7 T superconducting magnet providing a 34.7 T/m linear magnetic field gradient (the details of the pulse sequence can be found in Refs. [15,32,64]).

An Oxford 1200CF continuous flow cryostat was employed for the temperature measurements in the range between 100 K to 400 K. The accuracy of the temperature was ± 1.0 K. The samples were initially cooled down to 100 K, and the NMR measurements were obtained on a heating cycle. This was followed because it is well known that certain ILs exhibit a notable hysteresis behavior, for example, in their ionic conductivity [70] and in their NMR relaxation measurements [71,72].

3.3. Software

To optimize the structure of the molecules of the IL used in this study, the free and open-source software ORCA (Open-Source Computational Chemistry Software) was used. This software utilizes a combination of ab initio, density functional theory, and semi-empirical methods to perform calculations and optimization of molecular systems. It can handle a variety of molecular systems, ranging from small to large molecules. Full documentation about this software can be found in the original references [44,45].

4. Conclusions

Measurements were carried out over a wide range of temperatures (100 K–400 K), covering the bulk IL and confined IL dynamics below and above their glass transition points, 186.4 K and 190.5 K, respectively. In the high-temperature regime (above T_g), the dynamical behavior of the IL molecules depends on the molecular tumbling, reorientations of proton groups, and translational motions associated with self-diffusion. It was observed that, in the bulk ionic liquid, D follows a VFT temperature dependence indicative of a fragile liquid behavior towards the glass transition temperature. In this case, the underlying motion is predominately translational diffusion motion. On the other hand, the T_1 relaxation data also revealed re-orientational motions that predominately originated from the rotation of the two CH₃ groups of the IL molecules. These methyl rotations manifest themselves clearly by the appearance of the T_1 minimum in the low-temperature regime below T_g . In this regime, all translational degrees of freedom have frozen, while the methyl group rotation persists even below T_g .

The effect of confinement on the dynamic properties of the ionic liquid clearly manifests itself in both temperature regimes. In the high-temperature liquid state, the mobility of the confined cations reduces significantly compared to that of the bulk IL as revealed by the

lower values of the self-diffusion coefficients. The T_1 relaxation data are highly influenced by the presence of pore walls of the mesoporous materials, especially at high temperatures.

We believe that further research is warranted to extend the results of this study and suggest the following.

First, obtain additional experimental self-diffusion data points over a wider temperature range than what is presented in the current work. This will provide a definitive conclusion as to whether the temperature evolution of the diffusion rate of confined IL follows an ideal- or fragile-liquid behavior. Second, conduct further theoretical work, such as MD simulations, to investigate the dynamics of IL molecules on the silica surface of pure SBA-15, which only contains OH groups on its surface, and SBA-15 with impurities represented by metal ions versus temperature. This work could then provide deeper insight into the origins of the anomalous dynamics of confined IL compared to bulk IL, especially at high temperatures above the glass transition.

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Data Availability Statement: Not applicable.

Conflicts of Interest: The authors declare no conflict of interest.

Appendix A

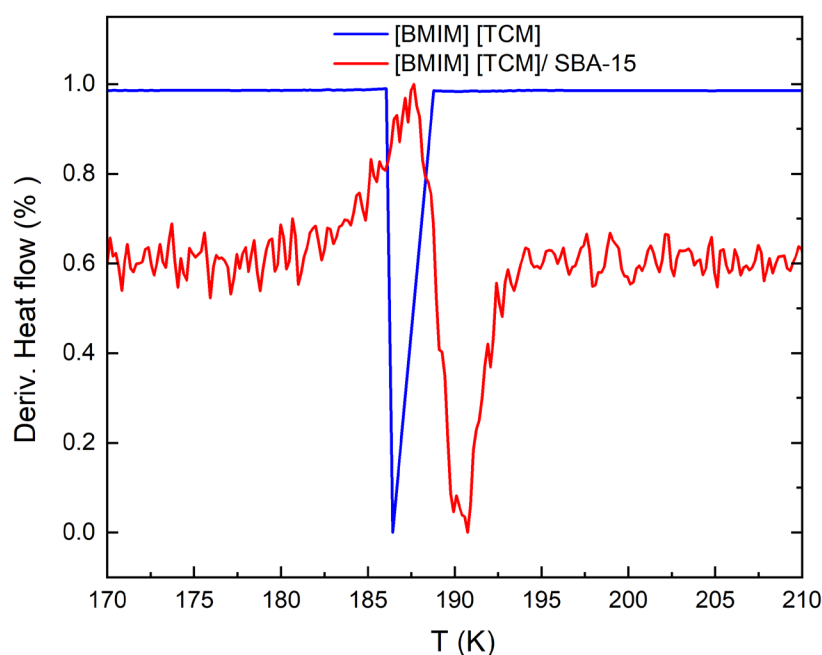


Figure A1. Results of the derivatives of the heat flow in the relevant Differential Scanning Calorimetry (DSC) experiments. As observed, there is a difference of approximately 4.1 K between the two glass transition temperatures (T_g for the bulk IL is 186.4 K and T_g for the confined IL inside SBA-15 is 190.5 K).

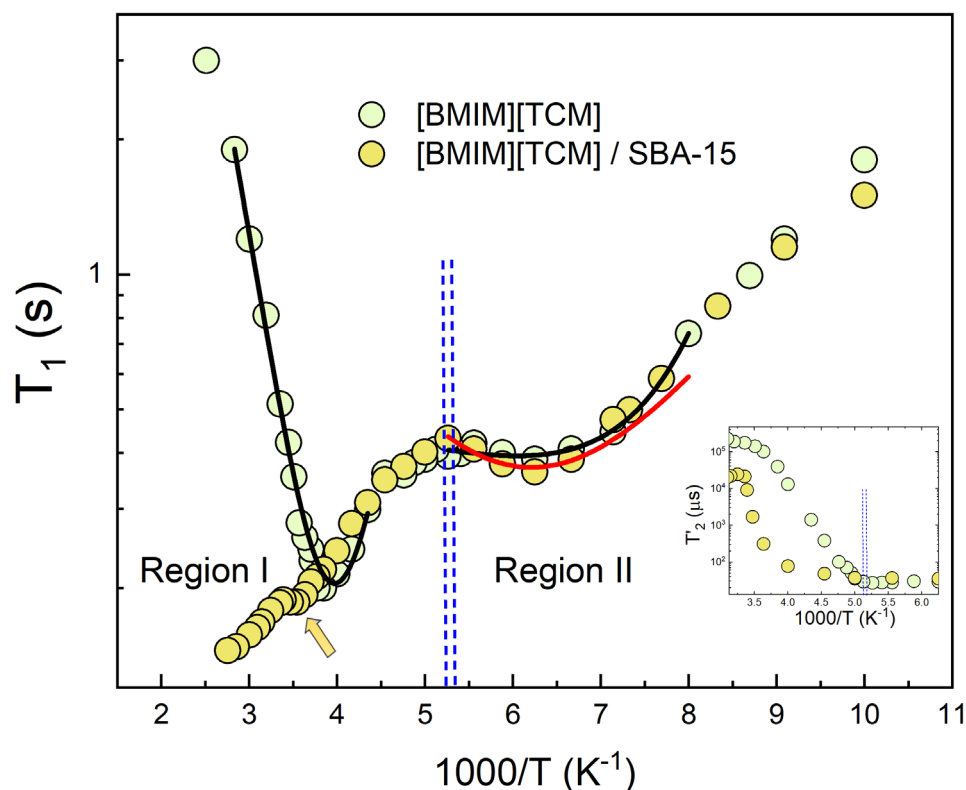


Figure A2. Replicate of the data shown in Figure 2 of the main text. Here, spin-relaxation times are shown versus reciprocal of temperature for both the bulk and confined ionic liquids. The lines are fit using the BPP model (refer to the main text for details). The two vertical dashed lines are the glass transition temperatures for the bulk ($T_g = 186.4$ K) and confined ILs ($T_g = 190.5$ K) obtained using the DSC technique (shown in Figure A1). The inset of the figure shows the spin-spin relaxation times T_2 vs. T .

Table A1. Statistical parameters obtained from fitting the self-diffusion data vs. $1000/T$ for the bulk and confined ILs using the Arrhenius and VFT models.

	Bulk IL		Confined IL	
	Arrhenius	VFT	Arrhenius	VFT
Reduced χ^2	5.9×10^{-22}	1.28×10^{-22}	8.2×10^{-25}	5.2×10^{-22}
Adjustable R^2	0.995	0.976	0.971	0.93
F-value	271	2343	18.6	63.9
RSS *	5.9×10^{-21}	1.4×10^{-21}	8.2×10^{-22}	1.03×10^{-21}

* RSS: Residual sum of squares.

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